Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/06/2020
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling Detached Bridgman Crystal Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2019
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:			
Flight Assignment:	NOTE: End date change to 12/31/2019 per NSSC info (Ed., 4/30/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Volz, Martin Ph.D. (NASA Marshall Space Flight Center) Croell, Arne Ph.D. (NASA Marshall Space Flight Center)		
Grant/Contract No.:	NNX16AC37G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The original goal of this research project was to study detached Bridgman growth in both microgravity and terrestrial environments using numerical models. One goal of this work was to promote collaboration between research groups at the University of Minnesota (UMN) and the NASA Marshall Space Flight Center (MSFC). A research team at MSFC was to focus on experimental aspects of detached Bridgman growth, while UMN researchers were to develop and apply sophisticated mathematical models to analyze the process. Toward developing tasks to promote these original objectives, Principal Investigator (PI) Derby visited Dr. Martin Volz at the NASA Marshall Space Flight Center on November 29, 2016, to discuss past progress and future research directions. However, after discussions with Dr. Volz and others, a shift in research focus was deemed appropriate, and this project has been redirected to address continuing opportunities for advancing the understanding of the mechanisms that determine the pushing or engulfment of foreign particles in a melt at a moving solid-liquid interface. Of particular interest and relevance is the engulfment of silicon carbide and silicon nitride particles during the solidification of multi-crystalline silicon used in photovoltaic devices. This work has and will continue to coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft und Raumfahrt) and involving the Fraunhofer IISB and the University of Freiburg. Dr. Martin Volz of Marshall Space Flight Center is also involved with this work on particle engulfment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A better understanding of particle engulfment is vital to improving the quality of solar-grade, multi-crystalline silicon that is grown via directional solidification processes. Understanding the dynamics of particle engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production. In addition, particle engulfment is also important in other physical processes, such as the fabrication of metal-matrix composites, separation processes, cryogenic preservation of biological materials, and frost heaving.
Task Progress:	This project was marked by several, significant milestones, starting with a detailed analysis and implementation of a computational model for particle engulfment during solidification. Subsequently, the model was applied to more fully understand the details of silicon carbide, SiC, engulfment during the growth of silicon. Our advances in knowledge were reported by many conference presentations and seminars, as well as a series of journal publications (these are listed at the end of this report and may be found in the Cumulative Bibliography hyperlink for this project), An important advance for this project was the discovery of a new scaling law that described the dependence of the critical velocity on the size of the particle for the engulfment of silicon carbide (SiC) particles during silicon solidification. Whereas all prior analytical models had predicted that the critical velocity scaled as either the inverse of particle radius or the radius raised to the -4/3 power, our finite element model showed a dependence on the radius raised to the -5/3 power. This finding was significant, since for the first time in over a decade of research on SiC inclusions in silicon, our model was able to provide a quantitative correlation with experimental results, and furthermore allowed for the unambiguous identification of the underlying physical mechanisms that gave rise to the observed behavior of this system. In particular, we identified a significant and previously unascertained interaction between particle-induced interface deflection (originating from the thermal conductivity of the SiC particle being larger than that of the surrounding silicon liquid) and curvature-induced changes in melting temperature arising from the Gibbs-Thomson effect. For a particular range of particle sizes, the Gibbs-Thomson effect flattens the deflected solidification interface, thereby reducing drag on the particle and increasing its critical velocity for engulfment. We showed via numerical calculations and analytical reasoning that these effects gave rise to the new scaling of the critical velocity to particle radius. The original work on this project involved the use of an in-house, research code to computationally model particle engulfment. While this code was powerful and efficient, it was unsuitable for general usage and distribution. Therefore, later effort involved the development and application of a new code, the open-source, finite-element code Goma 6.0, to describe engulfment physics. Goma 6.0 is a multiphysics finite-element code with a basis in computational fluid dynamics for problems with free and moving surfaces. It is therefore uniquely suited for the modeling of continuum transport and the representation of the many interfaces in the engulfment problem. Significant benefits are expected through the use of Goma 6.0 for this problem. Not only will this code provide a platform to continue to analyze the engulfment problem, but, importantly, the software produced will allow a broader user community to also use this tool to study engulfment phenomena. Finally, this project also supported one graduate student, Dr. Yuta Tao, who received his Ph.D. in materials science in December 2018. Subsequent work was carried on via a series of post-doctoral research associates, including Dr. Jeffrey H. Peterson, Dr. Benjamin Drueke, Dr. Jan Seebeck, and Dr. Chung-Husan Huang. Publications Derby JJ. "The synergy of modeling and novel experiments for melt crystal growth research." IOP Conf Ser: Mater Sci Eng. 2018;355:12001. https:// Derby JJ. "Fluid dynamics in crystal growth: The good, the bad, and the ugly." Progress in Crystal Growth and Characterization of Materials. 2016 Jun;62(2):286-301. (in Special Issue: Recent Progress on Fundamentals and Applications of Crystal Growth; The Proceedings of the 16th Summer School on Crystal Growth (ISSCG-16), Otsu, Shiga, Japan, August 1–7, 2016.) https:// Derby JJ, Tao Y, Reimann C, Friedrich J, Jauss T, Sorgenfrei T, Cröll A "A quantitative model with new scaling for silicon carbide particle engulfment during silicon crystal growth" J Crystal Growth 463, 100–109 (2017). https:// Tao Y, Sorgenfrei T, Jauss T, Cröll A, Reimann C, Friedrich J, Derby JJ "Particle engulfment dynamics under oscillating crystal growth conditions" J Crystal Growth 468, 24–27 (2017). https://

	Friedrich J, Reimann C, Jauss T, Cröll A, Sorgenfrei T, Tao Y, Derby JJ "Engulfment and pushing of Si3N4 and SiC particles during directional solidification of silicon under microgravity conditions" J Crystal Growth 475, 33–38 (2017). https://
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Derby JJ. "The synergy of modeling and novel experiments for melt crystal growth research." IOP Conf Ser: Mater Sci Eng. 2018;355:12001. https://doi.org/10.1088/1757-899X/355/1/012001 , Jan-2018
Articles in Peer-reviewed Journals	Derby JJ. "Fluid dynamics in crystal growth: The good, the bad, and the ugly." Progress in Crystal Growth and Characterization of Materials. 2016 Jun;62(2):286-301. (in Special Issue: Recent Progress on Fundamentals and Applications of Crystal Growth; The Proceedings of the 16th Summer School on Crystal Growth (ISSCG-16), Otsu, Shiga, Japan, August 1–7, 2016.) https://doi.org/10.1016/j.pcrysgrow.2016.04.015 , Jun-2016